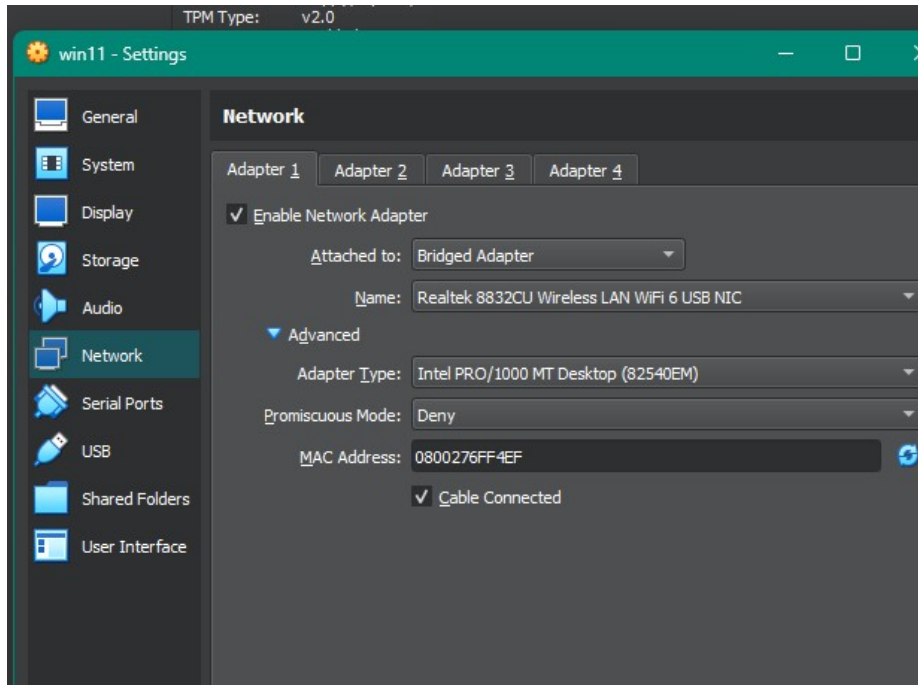


Windows 11 Networking and Troubleshooting

No internet due to bad DNS

Establish a baseline working state

1. Configure VM network mode to bridged adapter.



2. Verify IP address assignment and ping external address.

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\helpdesk>ipconfig /all

Windows IP Configuration

Host Name . . . . . : win11
Primary Dns Suffix . . . . . : mydomain.local
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : mydomain.local

Ethernet adapter Ethernet:

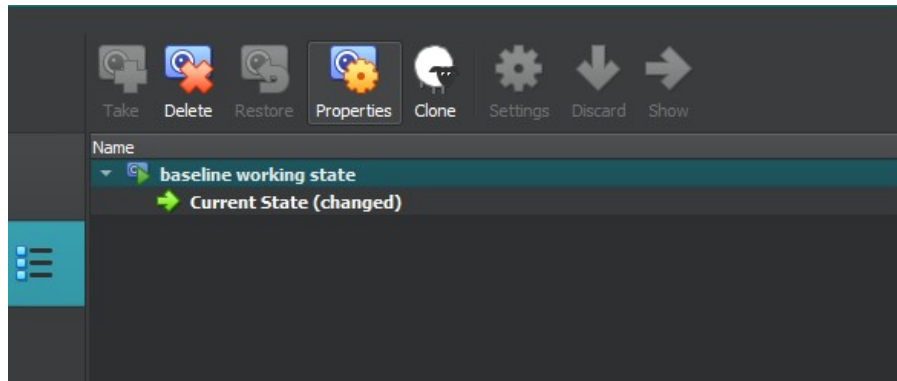
Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-6F-F4-EF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::3976:cbee:2ab6:48ce%9(Preferred)
IPv4 Address. . . . . : 192.168.40.37(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Wednesday, April 30, 1890 11:03:52 AM
Lease Expires . . . . . : Sunday, June 7, 2026 2:32:09 AM
Default Gateway . . . . . : 192.168.40.1
DHCP Server . . . . . : 192.168.40.1
DHCPv6 IAID . . . . . : 101187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-94-2E-CB-08-00-27-6F-F4-EF
DNS Servers . . . . . : 192.168.40.36
                        fdad:bbcc:ddee::1
```

```
C:\Users\helpdesk>ping www.google.com

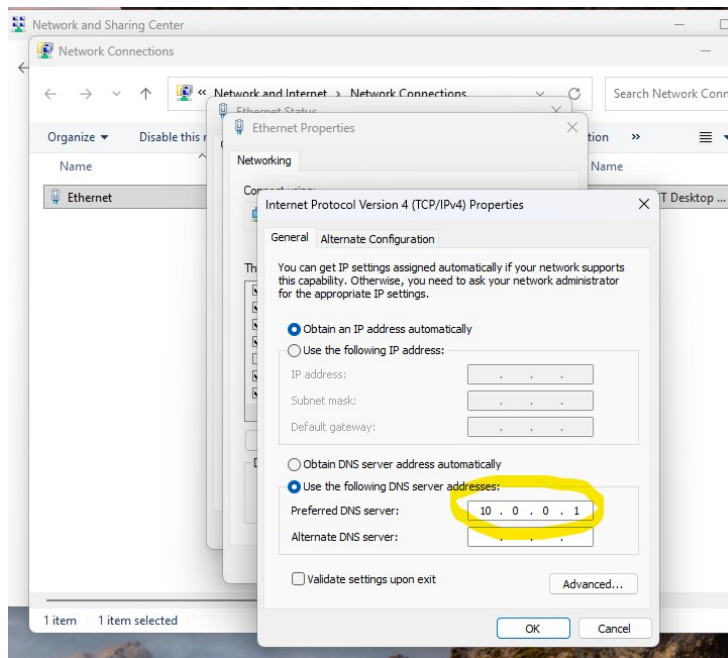
Pinging www.google.com [142.251.157.119] with 32 bytes of data:
Reply from 142.251.157.119: bytes=32 time=18ms TTL=118
Reply from 142.251.157.119: bytes=32 time=19ms TTL=118
Reply from 142.251.157.119: bytes=32 time=17ms TTL=118
Reply from 142.251.157.119: bytes=32 time=17ms TTL=118

Ping statistics for 142.251.157.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 19ms, Average = 17ms
```

(State saved in hypervisor)



Break the connection:



```
C:\Users\helpdesk>ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3976:cbee:2ab6:48ce%9
    Default Gateway . . . . . : 

C:\Users\helpdesk>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3976:cbee:2ab6:48ce%9
    IPv4 Address. . . . . : 192.168.40.37
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.40.1
```

```
C:\Users\helpdesk>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Users\helpdesk>
```

Broken state confirmed.

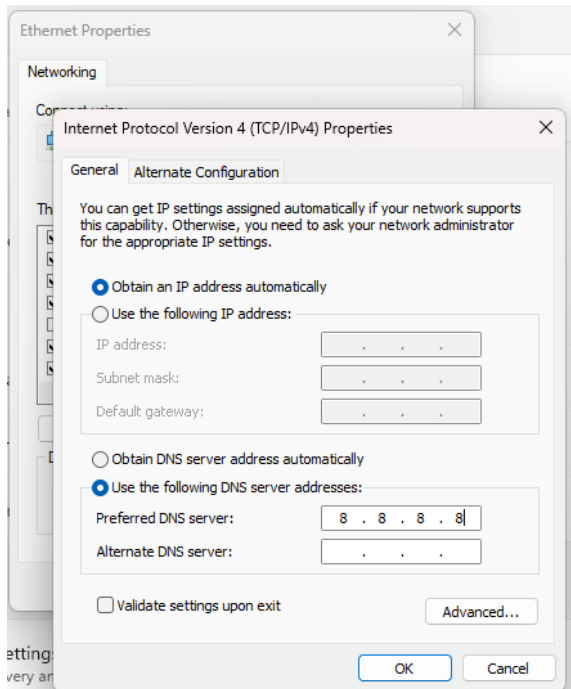
Ping DNS to confirm active connection.

```
C:\Users\helpdesk>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=20ms TTL=118
Reply from 8.8.8.8: bytes=32 time=18ms TTL=118
Reply from 8.8.8.8: bytes=32 time=18ms TTL=118
Reply from 8.8.8.8: bytes=32 time=16ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 20ms, Average = 18ms
```

Fix DNS server settings:



Flush DNS and ping external address:

```
C:\Users\helpdesk>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

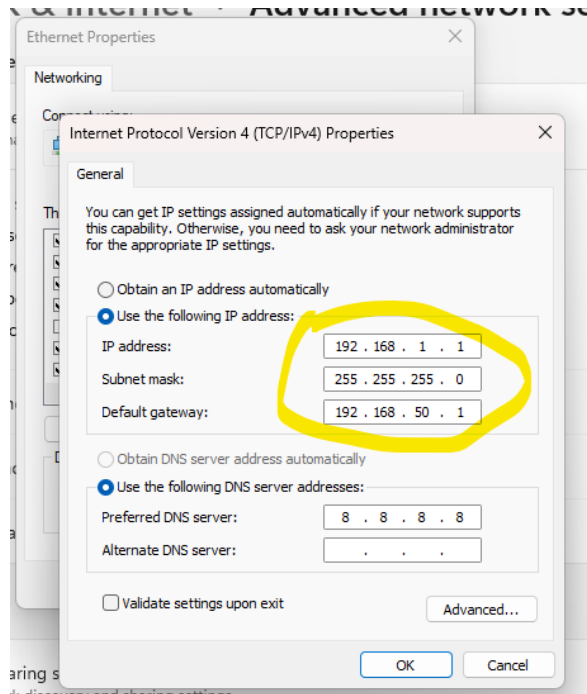
C:\Users\helpdesk>ping www.google.com

Pinging www.google.com [142.251.155.119] with 32 bytes of data:
Reply from 142.251.155.119: bytes=32 time=19ms TTL=118
Reply from 142.251.155.119: bytes=32 time=20ms TTL=118
Reply from 142.251.155.119: bytes=32 time=26ms TTL=118
Reply from 142.251.155.119: bytes=32 time=21ms TTL=118

Ping statistics for 142.251.155.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 19ms, Maximum = 26ms, Average = 21ms
```

No internet due to wrong static IP

Break the connection:



Try to ping the gateway:

```
C:\Users\helpdesk>ping 192.168.50.1

Pinging 192.168.50.1 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.1.1: Destination host unreachable.
Request timed out.

Ping statistics for 192.168.50.1:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
```

Ping the DNS server:

```
C:\Users\helpdesk>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Request timed out.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
```

```
C:\Users\helpdesk>tracert 8.8.8.8

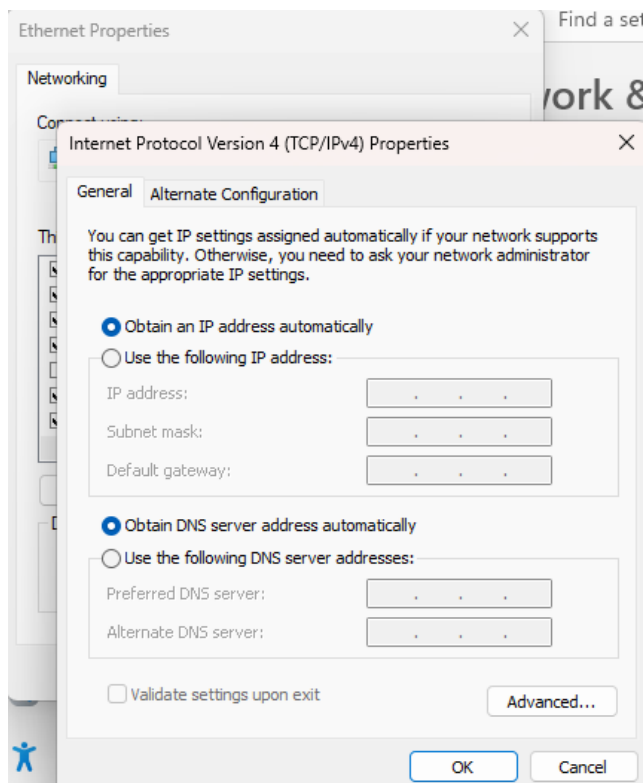
Tracing route to 8.8.8.8 over a maximum of 30 hops

 1    *    win11.mydomain.local [192.168.1.1]  reports: Destination host unreachable.

Trace complete.
```

No hops available.

Switch back to automatic DNS and dynamic IPV4 configuration:



Refresh the IP address:

```
C:\Users\helpdesk>ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3976:cbee:2ab6:48ce%9
    Default Gateway . . . . . : 

C:\Users\helpdesk>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3976:cbee:2ab6:48ce%9
    IPv4 Address. . . . . : 192.168.40.37
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.40.1
```

Test the configuration:

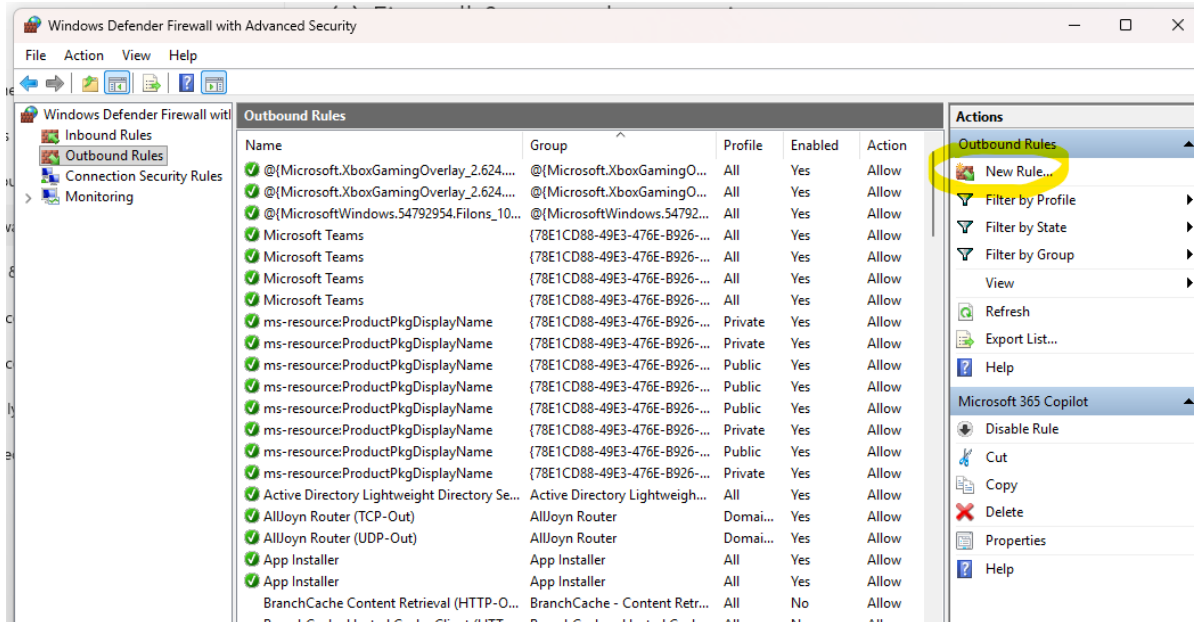
```
C:\Users\helpdesk>ping www.google.com

Pinging www.google.com [142.251.154.119] with 32 bytes of data:
Reply from 142.251.154.119: bytes=32 time=19ms TTL=118
Reply from 142.251.154.119: bytes=32 time=23ms TTL=118
Reply from 142.251.154.119: bytes=32 time=20ms TTL=118
Reply from 142.251.154.119: bytes=32 time=22ms TTL=118

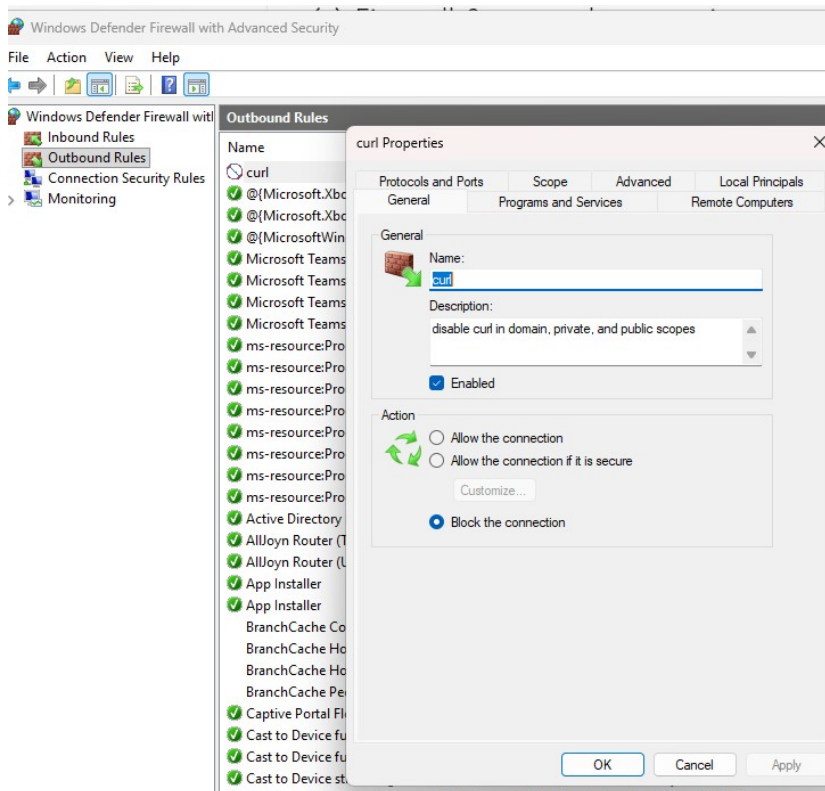
Ping statistics for 142.251.154.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 23ms, Average = 21ms
```

Blocked App due to Windows Defender Firewall rules

Opened windows defender firewall with advanced security. Created new rule.



Set to block the connection. Applied to the domain, private and public scopes.



Tested application:

```
C:\Users\helpdesk>curl www.google.com
curl: (7) Failed to connect to www.google.com port 80 after 65 ms: Could not connect to server
```


Deleted the rule. Tested the application a second time.

```
C:\Users\helpdesk>curl www.google.com
<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en"><head><meta content="Search the world's
information, including webpages, images, videos and more. Google has many special features to help you find exactly wha
t you're looking for." name="description"><meta content="noodp, " name="robots"><meta content="text/html; charset=UTF-8"
http-equiv="Content-Type"><meta content="/images/branding/googlelog/1x/googlelog_standard_color_128dp.png" itemprop="image"
><title>Google</title><script nonce="9aQhDFKxQkkT2FXQL0LFWA">(function(){var _g={kEI:'2KUkau6yLIOUruEP1NKk-Qw',kEXPI:'0,
1304203,2935842,14111,64702,360901,5532799,482,114,36810973,25228681,123988,28395,65165,138349,37800,107648,7707,7,33385
,51821,33266,13074,32560,6436,9743,2646,4534,19712,15888,1,2,1515,3355,196,5814,2,7421,12885,2,873,2,2230,5108,3008,13,7
82,4074,4,32545,5337,310,535,2046,2,11,2,8513,2,1746,4,3530,3,1345,5773,7326,1,968,2,147,389,4,2647,5,5568,298,6256,2997
,6032,5,10456,2,6335,95,1554,4,5749,1094,5833,2,4981,1,8,815,2,3768,4,1634,544,4,4000,187,2105,5,461,3925,4,2364,2363,5,
39,676,5743,4,2050,265,106,5,3,1039,902,5,4916,4,130,2573,4,40,4,1854,256,4,438,1137,1,1538,3096,4,1201,5,75,4,1940,4,14
47,4,198,4,1341,1,1357,2701,4,2902,1802,507,1813,41,1438,4,24,315,695,7,3356,1,3823,3449,4,480,3,21017368,4,2960,3,10055
,3,11871,6097,3,5578,1,6506615,6,5602,2,2628,70,1372,3901,1128,58,370,91901,144,2437572,1078330,1008832,12313199,2,69157
1,260789,34287,10,3,164821,3,2592126,5,3687,3,6,607,4,453,2,1167,4,489,61,1843,5,4928,7,7,7,865,126,5208,3964,1023,1,4,1
043,551,5,142,4,228,1621,2060,7,1240,582,4,523,1948,1973,5,2,7,960,26,1241,77,698,1536,60,2292,3641,262,485,5,8,1107,4,5
84,923,4,1628,2,3172,401,3566,4,99,4,3523,4,1740,2,280,3,456,5,225,539,4,553,222,1487,2660,1,2756,1201,5,1884,629,4,234,
1005,4,1348,734,1525,1,282,204,4,59,4,414,2425,5,47,4,1108,128,794,5,116,694,1,1,439,4,375,2395,1,10,188,637,1,406,4,120
```

Lessons learned:

1. Diagnosing network failures — Learned how to isolate connectivity issues using core tools (ipconfig, ping, tracert, nslookup) and interpret results to identify whether the failure was DNS, gateway, or IP-related.
2. Understanding IP configuration — Gained hands-on experience with DHCP vs. static addressing, subnet masks, default gateways, and how incorrect settings break communication.
3. Troubleshooting DNS problems — Practiced identifying name-resolution failures, testing raw IP connectivity, and correcting DNS server misconfigurations.
4. Using Windows Firewall for diagnostics — Learned how outbound rules can block applications, how to create and remove rules, and how to verify firewall-related connectivity issues.
5. Resetting and repairing network stacks — Demonstrated the ability to restore connectivity using ipconfig /release, ipconfig /renew, ipconfig /flushdns, and by resetting network adapters.